Adaptive Truncation and Simulation-Based Calibration Applied to COM-Poisson Models

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The Challenge of Count Data

Count data are ubiquitous across various scientific and social domains, representing the frequency of occurrence of discrete events.

- **Common Examples:**
  - Number of website visits.
  - Purchases in physical or online stores.
  - Calls to a call center.
  - Disease cases in a population.
  - Species abundance in a habitat.
- **Real-world data often exhibit:**
  - Equidispersion (Variance $=$ Mean)
  - Overdispersion (Variance $>$ Mean)
  - Underdispersion (Variance $<$ Mean)
- **Challenge:** Accurately modelling the inherent variability in these data.

- Hilbe, 2012
- Sellers, 2021
- Cameron & Trivedi, 2013

The Poisson Distribution: Fundamentals

The Poisson distribution has been the standard choice for modelling count data due to its simplicity and interpretability.

- **Probability Mass Function (PMF):**

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

  Where $\lambda > 0$ is the rate parameter.

- **Key Assumption: Equidispersion**
  - The mean and variance are equal: $E[X] = Var[X] = \lambda$.
  - This is a defining characteristic of the Poisson distribution.

▶ Shmueli et al., 2005

The Need for Flexible Models

To overcome the limitations of the Poisson distribution, especially its restrictive equidispersion assumption, more flexible models are needed.

- **Alternative Models:** Negative Binomial distribution (for over-dispersion), zero-inflated models, etc.
- **Challenge:** Finding a model that can uniformly accommodate both over-dispersion and under-dispersion.
- **The Path Forward:** The Conway-Maxwell-Poisson (COM-Poisson) distribution emerges as a promising solution.

▶ Hilbe, 2012

The Conway-Maxwell-Poisson (COM-Poisson) Distribution

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The COM-Poisson or CMP distribution generalises the Poisson by incorporating a dispersion parameter (ν), allowing it to model a wide range of dispersion patterns.

- **Origin:** Introduced by Conway and Maxwell (1962) in the context of queuing theory.
- **Flexibility:** Capable of capturing:
  - Under-dispersion ($Var[X] < E[X]$)
  - Equidispersion ($Var[X] = E[X]$, reducing to Poisson when $\nu = 1$)
  - Over-dispersion ($Var[X] > E[X]$)
- **Parameters:** $\lambda > 0$ (rate parameter) and $\nu > 0$ (dispersion parameter).
- **Usual reparametrisation:**
  - $\mu := \lambda^{1/\nu}$ (asymptotic approximation for $E[X]$), $\nu > 0$
  - Important for regression modelling

- Shmueli et al., 2005
- Conway & Maxwell, 1962

Dispersion Regimes of the COM-Poisson

The flexibility of the COM-Poisson is evident in its ability to adapt to different dispersion regimes, depending on the values of ν.

Parameters	Resulting Distribution	Dispersion Type
$\nu = 1$	Poisson	Equidispersion (Variance = Mean)
$\lambda < 1,\ \nu \to 0$	Geometric	Over-dispersion (Variance > Mean)
$\nu \to \infty$	Bernoulli	Under-dispersion (Variance < Mean)
$\nu > 1$	COM-Poisson	Under-dispersion (Variance < Mean)
$\nu < 1$	COM-Poisson	Over-dispersion (Variance > Mean)

Tabela: Dispersion regimes of the COM-Poisson distribution.

Probability Mass Function of the COM-Poisson

The PMF of the COM-Poisson is defined by two parameters, λ and ν, and includes a crucial normalising constant.

- **COM-Poisson PMF:**

$$P(X = k \mid \lambda, \nu) = \frac{\lambda^k}{(k!)^\nu Z(\lambda, \nu)}, \quad k = 0, 1, 2, \ldots$$

where

$$Z(\lambda, \nu) = \sum_{i=0}^{\infty} \frac{\lambda^i}{(i!)^\nu}$$

- **Problem:** $Z(\lambda, \nu)$ has no closed form (except when $\nu = 1$)

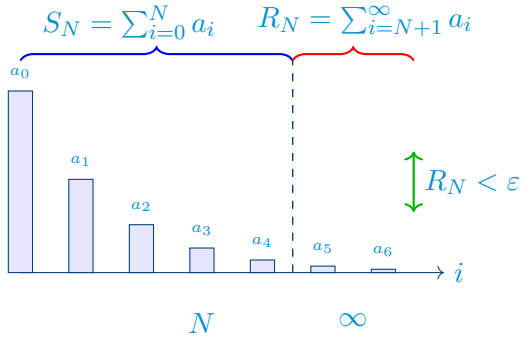
▶ Shmueli et al., 2005

Adaptive Truncation

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Adaptive truncation strategies provide a principled approach for approximating infinite sums with certified error bounds.

- **Core Idea:** Approximate an infinite sum by truncating it at a certain point, while providing a guarantee on the maximum error introduced by this truncation.
- **Benefit:** Allows for arbitrary precision (ε) in the computation of the normalising constant, crucial for robust statistical modelling.

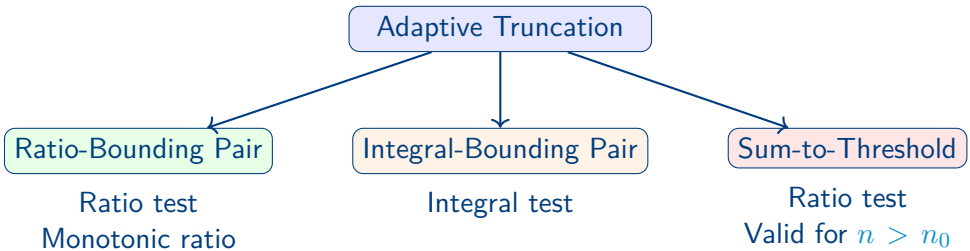


- Braden, 1992
- Carvalho, Silva, and Moreira, 2025

Adaptive Truncation

We propose 3 approaches, for a infinite sum $S = \sum_{n=0}^{\infty} a_n$:

- **Ratio-Bounding Pair**. Based on the ratio test for convergent series, providing explicit error bounds.
- **Integral-Bounding Pair**. Uses integrals to majorize and minorize the remainder of the series, suitable for functions that can be integrated, and has closed form.
- **Sum-to-threshold**. Truncates the series when the remaining terms fall below a certain threshold, only guaranteed to $n > n_0$, here n_0 can be very large.



- Braden, 1992
- Carvalho, Silva, and Moreira, 2025

Theorem 1: Ratio Bounding Pair dominates Sum-to-threshold

Let $\{a_n\}$ be positive, decreasing, and satisfy the ratio test with limit $L < 1$. If $r_n = \frac{a_{n+1}}{a_n}$ is monotonic, i.e., we can apply both approaches, Ratio Bounding Pair and Sum-to-Threshold. Then, the Ratio Bounding Pair approach uses fewer terms compared to the Sum-to-Threshold approach at the same precision level.

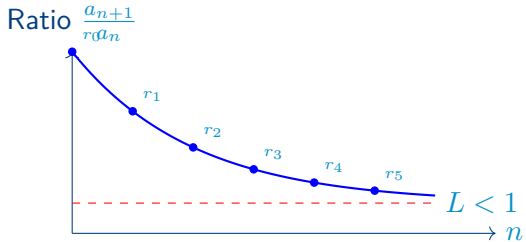
Proof of Theorem 1

See Appendix A.

Ratio Bounding Pair approach

The applicability of the Ratio Bounding Pair depends on the series satisfying the ratio test, which is a common condition for the convergence of many series.

- **Ratio Test Condition:** A positive series $\sum_{n=0}^{\infty} a_n$ converges if the limit of the ratio of consecutive terms is less than 1, i.e. $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = L < 1$.



The ratio of consecutive terms must converge to a value $L < 1$.

Ratio Bounding Pair approach

The main result formalises how to construct the error bounds for a series that satisfies the ratio test, providing the tools for adaptive truncation.

- **Context:** Given a series $S = \sum_{k=0}^{\infty} a_k$ with $a_k > 0$ that satisfies the ratio test with limit L.
- **Bounds:** If the ratio a_{n+1}/a_n is monotonically decreasing for L:

$$\frac{a_{n+1}}{1-L} < S - S_n < \frac{a_{n+1}}{1 - a_{n+1}/a_n}$$

If the ratio a_{n+1}/a_n is monotonically increasing for L:

$$\frac{a_{n+1}}{1 - a_{n+1}/a_n} < S - S_n < \frac{a_{n+1}}{1-L}$$

- **Implication:** These bounds allow estimating the error $S - S_n$ and ensuring it is within a tolerance ε.

▶ Carvalho, Silva, and Moreira, 2025

Ratio Bounding Pair for COM-Poisson

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The Ratio Bounding Pair approach is particularly suitable for the COM-Poisson normalising constant due to the structure of its terms.

- **COM-Poisson Series Terms:** The normalising constant $Z(\lambda, \nu) = \sum_{i=0}^{\infty} a_i$ is defined by the sequence $a_n = \frac{\lambda^n}{(n!)^\nu}, \quad n \geq 0.$
- **Ratio of Consecutive Terms:** The ratio between successive terms is

$$r_n = \frac{a_{n+1}}{a_n} = \frac{\lambda}{(n+1)^\nu}.$$

For $\nu > 0$, we have $r_n \to 0$ as $n \to \infty$, which guarantees the convergence of the series. Moreover, $(r_n)_{n \geq 0}$ is monotonically decreasing, ensuring that the conditions for the applicability of the Ratio-Bounding Pair method are satisfied.

- Sellers, 2021
- Conway & Maxwell, 1962
- Carvalho, Silva, and Moreira, 2025

Ratio Bounding Pair for COM-Poisson

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Algorithm: Ratio Bounding Pair

Initialize a_0, a_1 and $n = 1$

while $a_n > a_{n-1}$ **or** $a_n \left(1 - \frac{a_n}{a_{n-1}}\right)^{-1} \geq 2\varepsilon$ **do**

$n \leftarrow n + 1$

Evaluate a_n

end while

Evaluate $S_n = \sum_{k=0}^{n} a_k$

Return $S_n + \frac{a_n}{2}\left(\frac{L}{1-L} + \left(1 - \frac{a_n}{a_{n-1}}\right)^{-1}\right)$

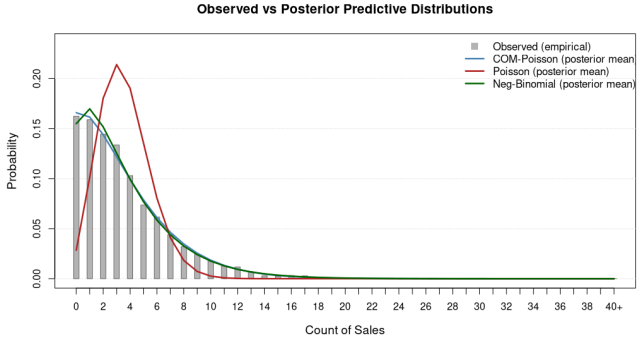
- In code, we use log-scale and logsumexp to prevent numerical errors.
- The normalisation constant of the COM-Poisson distribution starts increasing and from the term $\lfloor \lambda^{1/\nu} \rfloor$ it is decreasing.

▶ Carvalho, Silva, and Moreira, 2025

Case Study 1: inventory dataset

The utility of the COM-Poisson model is illustrated using real-world count data, as an alternative to the Poisson model.

- **Dataset:** Quarterly sales counts of a specific clothing item (with overdispersion).
- **Model Comparison:** The COM-Poisson provides the best visual fit, especially for low counts, outperforming the Negative Binomial. While the Poisson captures the mean, it fails to account for overdispersion and, thus, underestimates variability.



▶ Shmueli et al., 2005

Sensitivity to Numerical Precision

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How does the precision level affect sampling algorithms, like the MCMC?

Sensitivity to Numerical Precision

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How does the precision level affect sampling algorithms, like the MCMC?
We investigate how the numerical precision used to approximate the COM-Poisson normalising constant affects MCMC performance.

- **Key Component:**
  - The COM-Poisson PMF depends on $Z_\varepsilon(\lambda, \nu)$, computed numerically with precision ε
- **Experimental Design:**
  - 100 independent MCMC runs for each ε
  - Each run:
    - 1,000 warm-up iterations
    - 500 sampling iterations
- **Metrics**
  - Effective Sample Size (ESS)
  - Convergence diagnostic $\hat{R}$
  - Execution time and ESS/sec

Case Study 1: inventory dataset

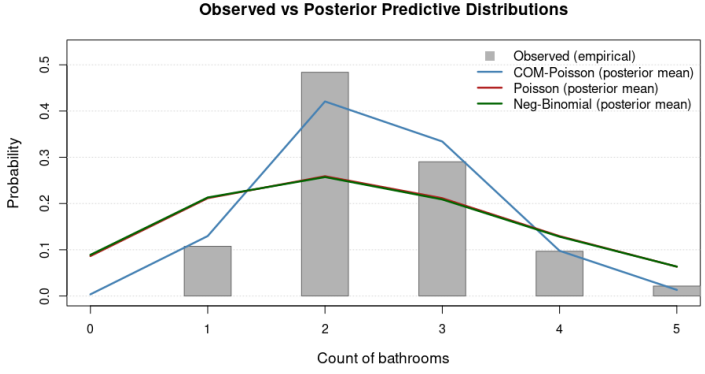
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- **Stable Convergence:** For high-precision settings ($\varepsilon = 2^{-32}, 2^{-16}, 2^{-8}$), MCMC convergence is stable.
- **Best Balance:** $\varepsilon = 2^{-8}$ provides good sampling efficiency and the lowest execution time among the high-precision settings.
- **Collapse at Low Precision:** At $\varepsilon = 2^{-4}$, convergence deteriorates severely, indicating that adequate numerical precision is essential for reliable inference in this highly over-dispersed dataset.

ε	ESS Mean	$\hat{R}$ Mean	Mean Time (sec)	ESS/sec
2^{-32}	93.6	1.016	1210.9	0.077
2^{-16}	**94.0**	**1.014**	776.3	0.121
2^{-8}	84.1	1.018	**522.9**	**0.163**
2^{-4}	5.5	1.577	22666.2	0.000

Case Study 2: house prices dataset

- **Dataset**: bathroom counts in residential properties (with underdispersion)
- **Model Comparison:** The COM-Poisson model provides the closest visual agreement with the empirical distribution of the bathroom count. In contrast, both the Poisson and Neg-Binomial models, while reaching the average of the distribution, fail to accurately fit the specific shape of the data.



Case Study 2: house prices dataset

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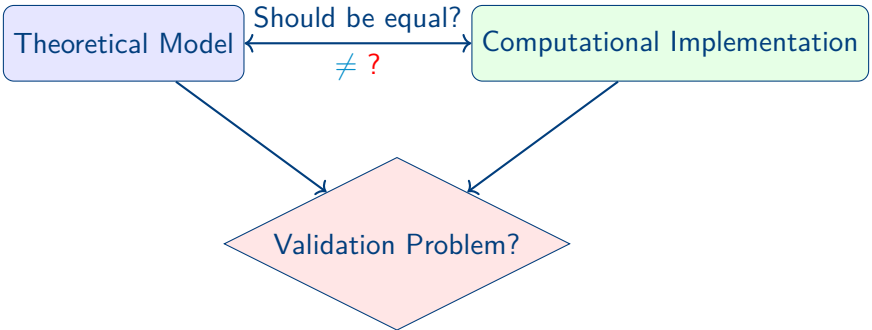
- **Stable Convergence**: For high-precision settings ($\varepsilon = 2^{-32}, 2^{-16}, 2^{-8}, 2^{-4}$) , MCMC convergence is stable.
- **Best Balance**: $\varepsilon = 2^{-4}$ provides the highest sampling efficiency and lower execution time among the high-precision settings.
- **Collapse at Low Precision**: At $\varepsilon = 2^{-1}, 2^{-2}$, convergence deteriorates severely.

ε	ESS mean	Rhat mean	Time mean (sec)	ESS/sec
2^{-32}	330.4	1.007	41.1	**8.047**
2^{-16}	323.1	1.006	**41.0**	7.872
2^{-8}	327.7	1.006	42.3	7.752
2^{-4}	**345.6**	**1.005**	43.3	7.990
2^{-2}	308.0	1.055	65.0	4.735
2^{-1}	263.4	1.155	101.0	2.609

The Challenge of Model Correctness

The validation of statistical inference procedures is crucial. Theoretical models and their computational implementations can diverge, leading to misleading results.

- **Discrepancies:** Can arise from numerical approximations, implementation errors, or unverified assumptions.
- **Impact:** Biased estimates, incorrect uncertainty quantification, and flawed scientific conclusions.



▶ Talts et al., 2020

SBC: A Framework for Validating Bayesian Inference

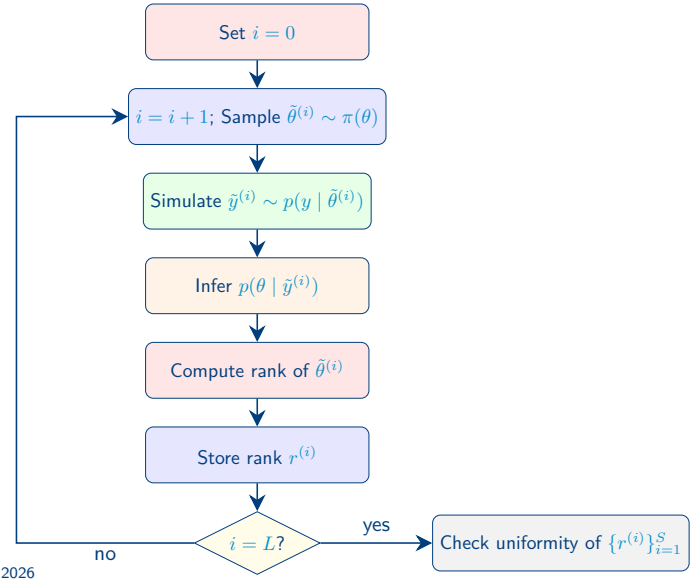
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Simulation-Based Calibration (SBC) is a robust methodology for validating Bayesian inference algorithms, ensuring that the posterior distributions are consistent with the generative model.

- **Central Principle:** If an MCMC sampler is correctly implemented and the model is valid, the distributions of the ranks of the true parameters within the posterior samples should be uniform.
- **Why it's Important:** It diagnoses systematic biases that traditional MCMC diagnostics (like $\hat{R}$ and ESS) may not detect.

- Talts et al., 2020
- Cook, Gelman, and Rubin, 2006

Simulation-Based Calibration (SBC) Flowchart



- Talts et al., 2020
- Michels's presentation, 2026

SBC Algorithm

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Algorithm: Simulation-Based Calibration Procedure

Require: Inference Algorithm $\mathcal{M}$ with prior $\pi(\theta)$ and likelihood $\pi(y \mid \theta)$; Number of simulations N; Number of posterior samples L.

for $i = 1$ to N **do**

Sample true parameter: $\tilde{\theta}^{(i)} \sim \pi(\theta)$

Simulate data: $\tilde{y}^{(i)} \sim \pi\left(y \mid \tilde{\theta}^{(i)}\right)$

Run inference algorithm to obtain L posterior samples: $\theta_1^{(i)}, \cdots, \theta_L^{(i)} \sim \mathcal{M}(\tilde{y}^{(i)})$

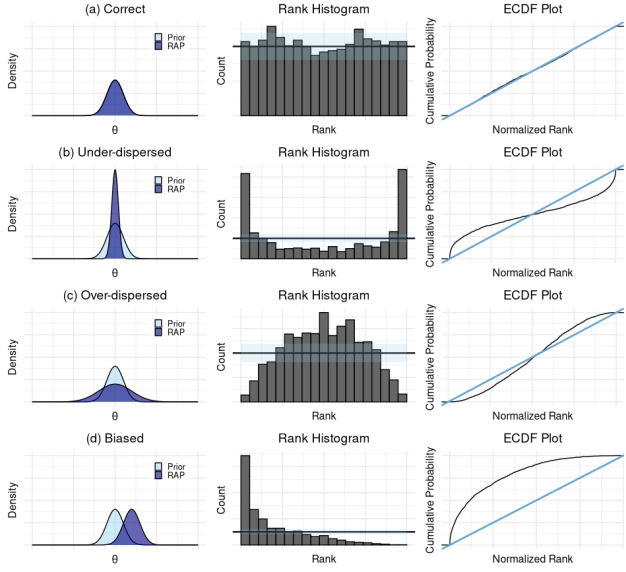
Calculate the rank: $r^{(i)} = \sum_{l=1}^{L} \mathbb{I}\left(\theta_l^{(i)} < \tilde{\theta}^{(i)}\right)$

end for

return $\left\{r^{(i)}\right\}_{i=1}^{N}$

▶ Talts et al., 2020

Simulation-Based Calibration (SBC)



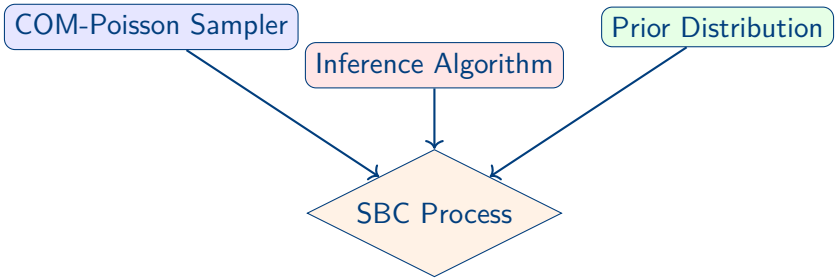
- Mendes et al., 2024

SBC for COM-Poisson Models: Requirements

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To perform Simulation-Based Calibration effectively for COM-Poisson models, several key components are required to ensure the validity of the calibration process.

- **Sampler for Data Generation:** A reliable sampler for generating data from the COM-Poisson distribution ($\tilde{y} \sim \pi(y \mid \tilde{\theta})$).
- **Well-Defined Prior:** A suitable prior distribution ($\pi(\theta)$) for the COM-Poisson parameters (μ, ν) (with reparametrisation $\mu := \lambda^{1/\nu}$) must be specified.
- **Inference Algorithm:** The Bayesian inference algorithm to be tested.



▶ Talts et al., 2020

Sampling from the COM-Poisson Distribution

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Sampling from the COM-Poisson distribution is essential for various tasks, including data simulation and Monte Carlo-based inference.

- **Ratio-of-Uniform Method:** A general method that allows sampling from a distribution $\pi(x) = \tilde{\pi}(x)/Z$ by sampling uniform variables.
- **Rejection Sampling:** Methods tailored for the COM-Poisson, often using Poisson or geometric distributions as envelopes.
- **Exact Simulation with Ratio Bounding Pair:** This method, adapted from the Ratio Bounding Pair approach, is particularly important as it ensures no approximation error in the data generation phase of SBC.

▶ Benson and Friel, 2021

Sampling from COM-Poisson Distribution

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Algorithm: Exact Simulation with Ratio Bounding Pair

Simulate $z \sim U(0,1)$
Initialize $k = 0$, $j = 0$
Compute S_j and S_k
Compute (L_k, U_k)
Compute $(\underline{F}_{j,k}, \overline{F}_{j,k})$
while $\underline{F}_{j,k} < z$ **do**
    if $\underline{F}_{j,k} < z < \overline{F}_{j,k}$ **then**
        Update $k = k + 1$
        Compute $a[k] = \tilde{\pi}(k)$
        Update $S_k = \mathrm{sum}(a)$
        Compute (L_k, U_k)
    else
        Update $j = j + 1$
        Update $S_j = \mathrm{sum}(a[1:j])$
    end if
    Compute $(\underline{F}_{j,k}, \overline{F}_{j,k})$
end while
Return j

Choice of Prior for the COM-Poisson

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Defining a suitable prior for the COM-Poisson parameters (μ, ν) (with reparametrisation $\mu := \lambda^{1/\nu}$) is crucial in Bayesian inference. In the context of Simulation-Based Calibration (SBC), the prior must be flexible and avoid overly informative assumptions.

- **In literature:** Conjugate priors have been proposed for the COM-Poisson model, but they are generally unsuitable for SBC. Existing works often suggest Gamma priors for both μ and ν.
- **Our choice:**
  - To balance under- and overdispersion: $\nu \sim \text{Gamma}(1, 1)$
  - For flexibility across a wide range of applications:

$$\mu \sim \text{Gamma}(\alpha, \beta), \quad \alpha, \beta > 0$$

  - In practice: $\mu \sim \text{Gamma}(0.1, 0.01)$. With Mean $= 10$ and Variance $= 1000$.

- Mikkola et al, 2024
- Benson and Friel, 2021
- Boatwright et al., 2006
- Meyer, Graye, and Sellers (2023)

SBC Experimental Configuration

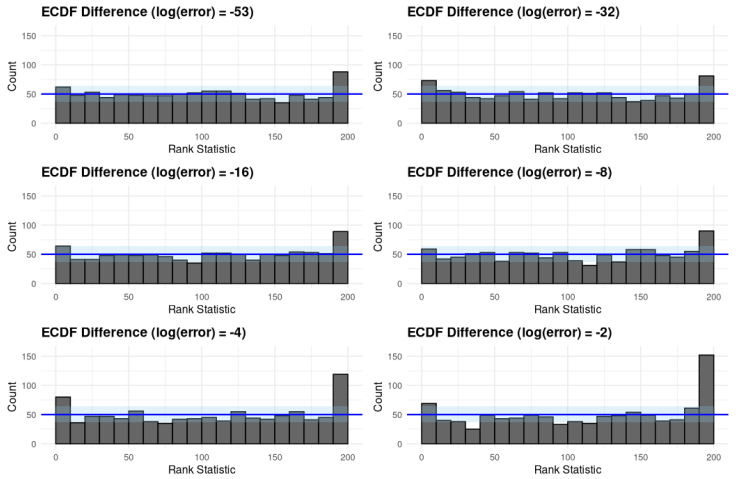
The Simulation-Based Calibration (SBC) experiments were meticulously designed to assess the computational faithfulness of the COM-Poisson inference algorithm across various precision levels.

- **Number of SBC Simulations (N):** 1,000 replications.
- **Number of Posterior Samples (L):** 200 posterior samples for rank calculation per replication.
- **Observations per Dataset:** 100 observations per simulated dataset.
- **Warmup Iterations:** 1,000 warmup iterations for the MCMC sampler.
- **Inference Algorithm:** MCMC implemented in Stan, utilising adaptive truncation for the COM-Poisson normalising constant.
- **Precision Levels (ε):** Six distinct levels of numerical precision were evaluated: $2^{-53}, 2^{-32}, 2^{-16}, 2^{-8}, 2^{-4}, 2^{-2}$.

▶ Talts et al., 2020

SBC Rank Histograms: Mean Parameter (μ)

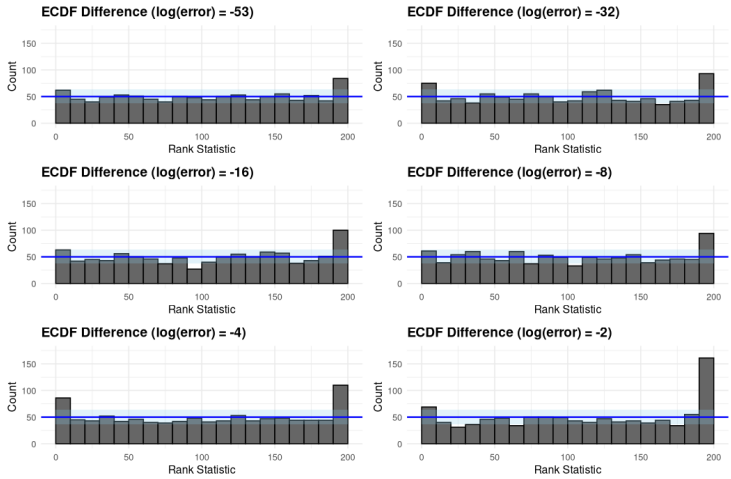
The rank histograms for the mean parameter (μ) visually demonstrate how numerical precision affects calibration.



SBC Rank Histograms: Dispersion Parameter (ν)

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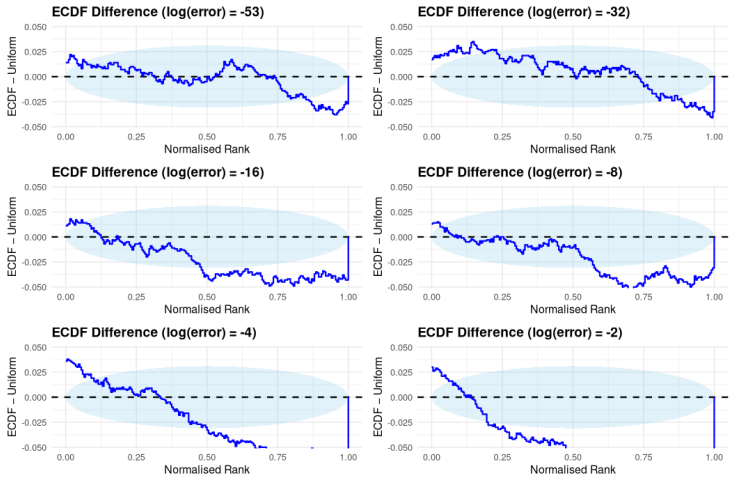
Similarly, the rank histograms for the dispersion parameter (ν) confirm the impact of precision on calibration.



ECDF Diagnostics: Mean Parameter (μ)

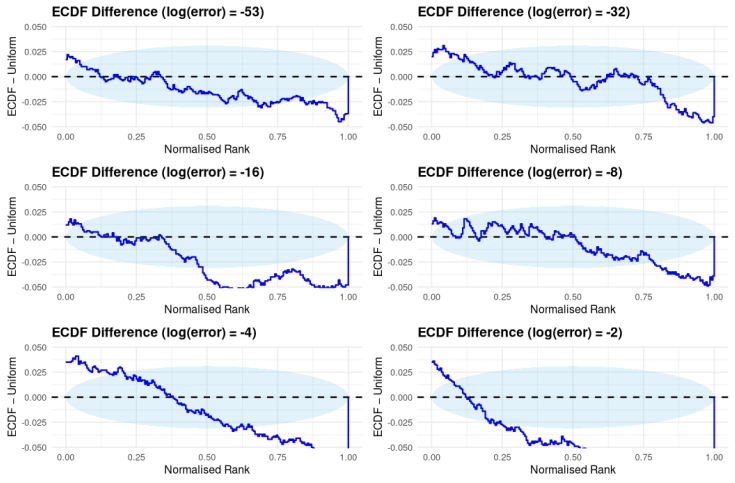
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The ECDF difference plots for μ provide a more granular view of the deviations from uniformity, complementing the rank histograms.



ECDF Diagnostics: Dispersion Parameter (ν)

The ECDF difference plots for ν reinforce the conclusions obtained for μ, highlighting the sensitivity of both parameters to numerical precision.



Future Work

- **Acceptance Probability Analysis**. Directly analysing how numerical errors in $Z(\lambda, \nu)$ propagate through the Metropolis-Hastings acceptance ratio, with the objective of understanding how these errors affect the behaviour of the MCMC algorithm.
- **Hybrid Approach**. Exploring hybrid methods combining asymptotic approximations with adaptive truncation.
- **Discrete Distribution Extensions**. Extending the adaptive truncation approach to a broader class of series could enable a wider range of applications.
- **Zero-Inflated Extensions**. Extending the computationally faithful COM-Poisson framework to include zero-inflated (ZICOMP) or hurdle variants to address datasets with excess zeros.
- **SBC Extensions**. Leveraging more advanced versions of SBC, such as Posterior SBC, to diagnose subtle forms of model misspecification or computational bias, particularly within hierarchical COM-Poisson models.

Thanks!

Any thoughts?

Special thanks to:

My supervisor (for patience!)
Coffee (for survival)
My family and friends (for everything)